

Program: Diploma in Civil Engineering and Planning	
Course Code: 6391A	Course Title: Basic Geotechnical Engineering
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge about the index and engineering properties of soil and soil classification
- To impart knowledge about soil exploration, bearing capacity and foundations in soil

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of mathematics		Engineering Mathematics	1
Basic knowledge about building components and construction techniques		Building Construction and Construction materials	3
Basics of mechanics		Engineering Mechanics	2

Course Outcomes:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Determine the physical and index properties of soil	18	Applying
CO2	Determine the engineering properties related to permeability and shear strength of soil.	12	Applying
CO3	Determine the engineering properties of soil related to compaction and consolidation	12	Understanding
CO4	Illustrate the process of field investigation and determine the bearing capacity of soil.	16	Applying
	Series tests	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3						
CO3							
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Determine the physical and index properties of soil		
M1.01	Understand the soil formation and scope of geotechnical engineering.	2	Remembering
M1.02	Solve the Functional relationships.	5	Applying
M1.03	Understand the procedures for determination of index properties of soil.	4	Understanding
M1.04	Understand Consistency of soils	3	Understanding
M1.05	Determine the gradation and classification of soils	4	Understanding

Contents:

Introduction to Soil Mechanics: Definition of soil, soil Engineering and Geotechnical engineering. Soil formation and transportation of soil - water transported soil, wind transported soil, glacier deposited soil and gravity deposited soil. Scope of geotechnical engineering.

Physical and Index Properties of Soil:

Soil as a three-phase system: Illustration of three phase system of soil mass. Define the terms - water content, void ratio, porosity, degree of saturation, percentage air voids, air content, density index, Unit weight of soil (bulk unit weight, dry unit weight, unit weight of solids, saturated unit weight, submerged unit weight). Specific gravity of solids.

Functional relationships: between void ratio and porosity; water content and void ratio; bulk unit weight and void ratio; dry unit weight and water content. Simple problems based on the above functional relationships.

Procedures for determination of: water content by oven drying method (BIS); field density of soil by core cutter and sand replacement method; specific gravity using pycnometer and specific gravity bottle

Consistency limits of soil: Procedure for determination of liquid limit, plastic limit and shrinkage limit. Define the terms - Plasticity index, consistency index, flow index and liquidity index. Gradation of soil: Define coarse grained and fine-grained soils. Procedure for determination of particle size distribution by sieve analysis and hydrometer analysis. Define effective diameter of soil, well graded soils, uniformly graded soils, Uniformity coefficient and coefficient of curvature. IS classification of soils			
CO2	Determine the engineering properties related to permeability and shear strength of soil.		
M2.01	Understand Soil water	1	Understanding
M2.02	Apply Darcy's law to find the co-efficient of permeability	4	Applying
M2.03	Outline the concept of shear strength of soil	4	Understanding
M2.04	Discuss the procedures of direct shear test and vane shear test	3	Applying
	Series Test – I	1	
Contents: Soil water: Types - Effective stress - Total stress - Pore pressure Permeability of Soil: Define permeability. State Darcy's law of permeability. List the factors affecting permeability. Procedure for determination of coefficient of permeability by constant head and variable head tests. Simple problems to find permeability of soil. Shear Strength of Soil: Define cohesion and internal friction. Define purely cohesive soils and cohesionless soils. Define shear strength of soil. Describe Mohr-Coulomb failure theory - Strength envelope, strength equation for purely cohesive and cohesion less soils. Procedure for determination of shear strength by direct shear test and vane shear test (simple problems on vane shear test only).			
CO3	Determine the engineering properties of soil related to compaction and consolidation		
M3.01	Understand the compaction of soil and procedure of Standard and Modified Proctor test	2	Understanding
M3.02	Explain the features of compaction curve and discuss the various field compaction methods	3	Understanding
M3.04	Outline the concept of consolidation of soil	3	Understanding
M3.05	Discuss the procedure of consolidation test. Discuss test results and basic definitions	4	Understanding

Contents:

Compaction of Soil: Define compaction of soil. Procedure for Standard and Modified Proctor Test.

Compaction curve - general discussion, differences of curves for sand and clay, Optimum moisture content (OMC), maximum dry density (MDD) and Zero air voids line. List the factors affecting compaction. Field methods of compaction - rolling, ramming and vibration. Suitability of different types of rollers - smooth wheel roller, sheep foot roller, pneumatic tyred roller.

Consolidation of soil: Define consolidation of soil- Difference between consolidation and compaction- different stages of consolidation (Initial, primary and secondary consolidation). Describe Terzaghi's soil-spring analogy model to explain the process of consolidation.

Consolidation test-consolidation test results- e-log p curve- Concepts of Coefficient of compressibility and volume compressibility- Compression index, Recompression index and normally consolidated, over consolidated and under consolidated soils

CO4	Illustrate the process of field investigation and determine the bearing capacity of soil.		
M4.01	Understand the basic concept of soil exploration	4	Understanding
M4.02	Describe the various methods of the explorations and define soil samples and samplers	4	Understanding
M4.03	Understand the basic concept of bearing capacity of a soil	4	Understanding
M4.04	Comprehend the type of foundation to be adopted for a particular soil condition	4	Understanding
	Series Test – II	1	

Contents:**Site investigation and soil exploration:**

List the objectives of site investigation. Discuss the different stages of site investigation. Criteria for deciding the depth, location and number of test pits or bore holes.

Discuss the different methods of soil exploration - Open excavation methods, different types of boring (Auger boring, wash boring, rotary drilling, percussion drilling, core drilling) and geo-physical methods. Discuss types of soil samples-list the types of soil samplers.

Bearing Capacity of Soil:

Define terms- bearing capacity, ultimate bearing capacity, safe bearing capacity and allowable bearing pressure. Types of bearing capacity failures. State Terzaghi's bearing capacity theory for strip footing [no derivation required] – Assumptions – Bearing capacity factors. Procedure for determination of bearing capacity of soil - Plate load test and Standard Penetration Test

Foundation: Different types of foundation - Distinguish shallow and deep foundation - different types of shallow foundations with sketches- classifications of pile foundation- Define well foundation- component parts of well foundation

Text / Reference:

T/R	Book Title/Author
T1	Gopal Ranjan & A.S.R Rao, ' <i>Basic and Applied Soil Mechanics</i> ', New Age International
R1	Punmia, B.C., ' <i>Soil Mechanics and Foundation Engineering</i> ', Laxmi Publication, Delhi.
R2	Arora K.R, ' <i>Soil Mechanics and Foundation Engineering</i> ', Standard Publisher.
R3	Murthy, V.N.S., ' <i>A Text book of Soil Mechanics and Foundation Engineering</i> ', CBS Publishers & Distributors Pvt. Ltd., New Delhi.
R4	Ramamurthy, T.N. &Sitharam, T.G., ' <i>Geotechnical Engineering (Soil Mechanics)</i> ', S Chand and Company LTD., New Delhi.
R5	Raj, P. Purushothama, ' <i>Soil Mechanics and Foundation Engineering</i> ', Pearson India, New Delhi.
R6	Kasamalkar, B. J., ' <i>Geotechnical Engineering</i> ', Pune Vidyarthi GrihaPrakashan, Pune.

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/105/103/105103097/
2	https://nptel.ac.in/courses/105/101/105101084/
3	https://www.astm.org/Standards/geotechnical-engineering-standards.html